Types of Data Models Exercises

1. What does a logical data model describe?

A. The physical location of files on a computer
B. How data is organized and how different pieces of data are related
C. The appearance of an application's interface
D. The hardware required to run a database

2. Which question does a logical data model help answer?

A. Which indexes should be created to improve query performance?
B. How should database files be distributed across storage devices?
C. How much disk space should be allocated to the database?
D. What rules govern relationships between records?

3. Which list identifies the three major logical data models?

A. Hierarchical, network, and relational
B. Physical, virtual, and external
C. Spreadsheet, document, and presentation
D. Sequential, alphabetical, and numerical

4. What basic structure does the hierarchical model use?

A. A collection of unrelated tables
B. A graph allowing multiple parents for each child
C. A tree of parent-child relationships
D. A single row of values

5. What is the topmost node in a hierarchical model called?

A. A child node
B. The root node
C. A foreign key
D. A relation

6. Which statement correctly describes parent-child relationships in a hierarchical model?

A. Each non-root node has one parent, and a parent may have many children.
B. Every child must have multiple parents.
C. Every parent must have exactly one child.
D. Nodes cannot have children.

7. What is a key difference between the network and hierarchical models?

A. The network model cannot represent relationships.
B. The hierarchical model allows every child to have multiple parents.
C. The network model organizes all data into tables.
D. The network model allows a record to connect to multiple parent records.

8. Which situation is better suited to the network model than to a strict hierarchical model?

A. Every record belongs to exactly one parent.
B. Records have no relationships.
C. A record needs connections to several parent records.
D. Data must form a single tree.

9. How does the relational model organize data and establish relationships?

A. It stores data in trees connected only by branches.
B. It organizes data into tables related primarily through keys.
C. It stores unrelated records without identifiers.
D. It organizes data according to physical disk locations.

10. How are records organized in the network data model?

A. In a tree where every child has only one parent
B. In separate files without relationships
C. As records connected through links in a network or graph
D. According to their physical storage locations